JOHN LEIGA

209.822.9104 | johnleiga9@gmail.com | LinkedIn

St. Louis, MO (Open to Relocation) | U.S. Citizen | Eligible to work on export-controlled programs

SUMMARY

Mechanical engineer focused on designing, building, and improving high-performance hardware systems under real-world constraints. Experience spans aerospace instrumentation and high-volume manufacturing systems, with ownership from concept through prototype, test, and production. Strong in first-principles problem solving, rapid iteration, and translating complex requirements into manufacturable hardware.

CORE SKILLS

Design & Analysis: Precision mechanical design, GD&T (ASME Y14.5), tolerance stack-ups, first-principles stress/deflection, structural FEA, DFM/DFA
Testing & Validation: Environmental qualification (vibration, shock, thermal), test planning & execution, instrumentation, failure analysis
Systems & Manufacturing: Mechanical-controls integration, root cause analysis, process optimization, capital equipment execution
Tools: SolidWorks, Autodesk Inventor, Python, MATLAB, Minitab
Hands-On: Prototype build, assembly, and test of mechanical systems

PROFESSIONAL EXPERIENCE

NESTLÉ PURINA PETCARE COMPANY — TECHNICAL APPLICATIONS GROUP
DESIGN ENGINEER | ST. LOUIS, MO | FEBRUARY 2025 – PRESENT

- Drive mechanical redesign and standardization efforts across multiple facilities to improve system performance and reliability
- Own mechanical specifications and operating standards across multiple production facilities
- Design extrusion dies and tooling; redesign and troubleshoot mechanical systems in active production environments
- Perform DOE and capability analysis to validate process improvements

NESTLÉ PURINA PETCARE COMPANY — MANAGEMENT DEVELOPMENT PROGRAM
MANAGEMENT DEVELOPMENT ENGINEER | BLOOMFIELD, MO & FORT DODGE, IA | MAY 2023 – FEBRUARY 2025

- Redesigned and stabilized a 14-furnace perlite expansion system through mechanical and controls-focused improvements
- Identified mechanical bottleneck (undersized rotary airlock) causing backpressure and shutdown events
- Implemented density-based feedback control and tuned system loops to improve stability and reduce manual intervention
- Designed and deployed zero-waste automated startup protocol; delivered **~$200K annual savings**
- Scoped and executed **~$500K in capital upgrades**, including vendor coordination and commissioning

TAVIS CORPORATION
PROJECT ENGINEER | MARIPOSA, CA | SEPTEMBER 2021 – MAY 2023

Re-Architected Aerospace Pressure Transducer (Saved $40M+ Program)

- Redesigned diaphragm-based sensing architecture using first-principles analysis to eliminate secondary reference system
- Reduced mechanical component cost ~50% and enabled ~30% system cost reduction
- Designed, built, and tested prototypes across full pressure range to validate performance
- Delivered full CAD assemblies, GD&T drawings, and supported manufacturing integration

Hardware Design, Modeling, & Qualification

- Built numerical modeling tool to predict nonlinear diaphragm behavior and sensor-gap sensitivity where FEA was insufficient
- Owned $10M+ aerospace hardware programs from requirements through qualification
- Executed environmental testing (vibration, shock, thermal) and correlated results with analytical predictions

INDEPENDENT PROJECTS

QUANTITATIVE RESEARCH & MODELING SYSTEM

- Built end-to-end Python system for data ingestion, feature engineering, model training, and backtesting
- Designed modular, CLI-driven architecture with reproducible experiment configuration
- Implemented multiprocessing workflows and large-scale data handling
- Developed model evaluation and explainability tools (SHAP)
- Designed and deployed full system architecture independently, demonstrating strong end-to-end engineering ownership

EDUCATION

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING
University of California, Merced — 2021